BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, BAHAWALPUR

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

MATHEMATICS

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Paper II (Objective Type) — Inter - A - 2022

Time Allowed: 30 Minutes | **Maximum Marks:** 20 | **Paper Code:** 8197

Note: Four possible choices A, B, C, D to each question are given. Which choice is correct fill that circle in front of that Question No. Use Marker or Pen to fill the circles. Cutting or filling two or more circles will result in Zero Mark in that Question.

Q1. Lim n -> infinity (1 - 1/n)^n = :

(A) e^2
(B) e
(C) e^-1
(D) -1/e

Q2. If f(x) = 1/(x-1) then f(2) = :

(A) -1
(B) 1
(C) 0
(D) -2

Q3. d/dx (y^n) = :

(A) ny^(n-1)
(B) ny^(n+1)
(C) ny^(n-1) dy/dx
(D) ny^(n-1) dx/dy

Q4. Lim x -> 0 ((e^x - 1)/x) = :

(A) 1
(B) 0
(C) e
(D) infinity

Q5. d/dx Cotax = :

(A) Cosec^2 ax
(B) aCosec^2 ax
(C) -aCosec^2 ax
(D) -aCosec ax

Q6. Integral of 3^(lambdax) dx = :

(A) 3^(lambdax)/ln3
(B) 3^(lambdax) ln3
(C) 1/lambda 3^(lambdax)/ln3
(D) 3^(lambdax)

Q7. Integral of dx/(a^2 + x^2) = :

(A) 1/a Tan^-1 x/a + C
(B) 1/(2a) ln (x-a)/(x+a)
(C) 1/(2a) ln (x+a)/(x-a)
(D) 1/a ln (a^2 + x^2)

Q8. d/dx (3^x) = :

(A) 3^x ln3
(B) 3^x
(C) x3^(x-1)
(D) 3^(x+1)

Q9. Integral of e^x(Cos x + Sin x)dx = :

(A) -e^x Sin x
(B) e^x Cos x
(C) -e^x Cos x
(D) e^x Sin x

Q10. Slope of the line ax + by + c = 0 is:

(A) a/b
(B) -b/a
(C) -a/b
(D) a/c

Q11. Distance of the point (2, 3) from y-axis is:

(A) 4
(B) 2
(C) -2
(D) 5

Q12. Integral from -pi to pi of Sinx dx = :

(A) 0
(B) 9
(C) 1
(D) 2

Q13. Equation of Horizontal Line through (7, -9) is:

(A) y = -9
(B) y = 7
(C) x = -9
(D) x = 7

Q14. A Circle is called a point circle if:

(A) r = 1
(B) r = 0
(C) r = 2
(D) r = 3

Q15. If m1 and m2 are the slopes of two lines, then lines are perpendicular if:

(A) m1m2 = 0
(B) m1m2 + 1 = 0
(C) m1m2 + 2 = 0
(D) m1 = m2

Q16. The point (-1, 2) satisfied the inequality:

(A) x - y > 4
(B) x - y >= 4
(C) x + y < 4
(D) x + y > 5

Q17. The length of Diameter of circle x^2 + y^2 = 9 is:

(A) 6 (B) 3

(C) sqrt(3) (D) 9

Q18. The non-zero vectors a and b are parallel if a x b = :

(A) 1 (B) -1

(C) 0 (D) 2

Q19. j x k = :

(A) i (B) -i

(C) 0 (D) k

Q20. Eccentricity of an Ellipse is:

(A) e = 0 (B) e > 1

(C) 0 < e < 1 (D) e = 1

Mathematics (Subjective) — Inter - A - 2022

Time Allowed: 2:30 Hours | **Maximum Marks:** 80

Note: Note: It is compulsory to attempt any(8-8)Parts each from Q.No. 2 and Q.No.3 while attempt any (9) Parts from Q.No.4. Attempt any (3) Questions from Part - II. Write same Question No. and its Part No. as given in the Question Paper — Part - I — 25 x 2 = 50

Q2.

Q.No.2

(i) Express the volume V of a cube as a function of the area of its base.

(ii) If f(x) = |x - 5| then find Left Hand and Right Hand limits of f(x) at x = 5

(iii) Find Domain and Range of f^-1 without finding inverse of f(x) if f(x) = (x-1)/(x-4), x not equal to 4

(iv) Find f(x) if f(x) = x^2 ln 1/x

(v) Find the limit Lim x -> 0 (1 - Cos2x)/x^2

(vi) Differentiate (a+x)/(a-x) w.r.t. x

(vii) If x = at^2 and y = 2at then find dy/dx

(viii) Find the Lim x -> 2 (x^3 - 8)/(x^2 + x - 6)

(ix) If y^3 + 3ax^2 + x^3 = 0, then find y2.

(x) Define Critical Value and Critical Point.

(xi) Find derivative of (x^3 + 1)^9 w.r.t. x

(xii) Differentiate 1/a Sin^-1 a/x w.r.t. x

Q3.

Q.No.3

(i) Use differential to approximate the value of Cos29 degrees.

(ii) Find delta y and dy of y = x^2 - 1 when x changes from 3 to 3.02.

(iii) Using differentials find dy/dx and dx/dy of xy - lnx = c

(iv) Solve the differential equation (e^x + e^-x) dy/dx = e^x - e^-x

(v) Evaluate integral of (1 - x^2)/(1 + x^2) dx

(vi) Evaluate integral of x/sqrt(4 + x^2) dx

(vii) Evaluate integral from 1 to 2 of (x^2 + 1)dx

(viii) Evaluate integral from 0 to pi/3 of Cos^2 theta Sin theta d theta

(ix) Find an equation of line through (-5,-3), (9,-1)

(x) By means of slopes, show that the points (-1,-3), (1,5), (2,9) lie on the same line.

(xi) Find the distance from the point P(6,-1) to the line 6x - 4y + 9 = 0

(xii) Find the lines represented by 2x^2 + 3xy - 5y^2 = 0

Q4.

Q.No.4

(i) Define an Objective Function.

(ii) Graph the solution set of Linear Inequality in xy-plane: 3y - 4 <= 0

(iii) Find the equation of circle with centre (sqrt(2), -3sqrt(3)) and Radius 2sqrt(2)

(iv) Find the Centre and Radius of a circle x^2 + y^2 + 12x - 10y = 0

(v) Find the equation of Tangent to the circle x^2 + y^2 = 25 at point (4,3)

(vi) Find Focus and Directrix of Parabola x^2 = 4(y-1)

(vii) Find Centre and vertices of the ellipse 9x^2 + y^2 = 18

(viii) Find the Eccentricity and Foci of Hyperbola y^2/16 - x^2/49 = 1

(ix) Write the vector PQ in the form of xi + yj when P = (2,3), Q = (6,-2)

(x) Find alpha so that |alpha i + (alpha+1)j + 2k| = 3

(xi) Find the Direction Cosines of the vector v = 3i - j + 2k

(xii) Find the Cosine of Angle between u = [2,-3,1], v = [2,4,1]

(xiii) Find the value of 3j . k x i

> Note: (Part - II) — 10 x 3 = 30

Q5. (Marks 10) (a) Evaluate Lim x -> 0 x/tanx (5) (b) If y = sqrt(x) - 1/sqrt(x), show that 2x dy/dx + y = 2sqrt(x) (5)

Q6. (Marks 10) (a) Evaluate integral of e^x ((1 + Sinx)/(1 + Cosx)) dx (5) (b) Find equations of two parallel lines perpendicular to 2x - y + 3 = 0 such that the product of the x and y intercepts of each is 3. (5)

Q7. (Marks 10) (a) Evaluate integral from pi/6 to pi/2 of Cosx/(Sinx(2 + Sinx)) dx (5) (b) Minimize z = 3x + y; subject to the constraints 3x + 5y >= 15; x + 6y >= 9; x >= 0; y >= 0 (5)

Q8. (Marks 10) (a) Find a joint equation of the straight lines through the origin and perpendicular to the lines represented by x^2 + xy - 6y^2 = 0 (5) (b) Show that the circles x^2 + y^2 + 2x - 2y - 7 = 0 and x^2 + y^2 - 6x + 4y + 9 = 0 touch externally (5)

Q9. (Marks 10) (a) Find Equations of the Tangents of the Ellipse x^2/128 + y^2/18 = 1 which are parallel to the line 3x + 8y + 1 = 0. Also find the points of Contact. (5) (b) Prove that Sin(alpha + beta) = Sinalpha Cosbeta + Cosalpha Sinbeta (5)